249438: porphyritic microgranite, Bannockburn mine

(Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Leonora (SH 51-1), Wildara (3041)
MGA Zone 51, 293398E 6850657N

Sampled on 1 September 2020

This sample was collected from the 270.91 – 272.91 m depth interval of drillcore BBDD0002, drilled in 2018 by Saracen Metals Pty Ltd at their Bannockburn prospect (Urquhart et al., 2019), and co-funded by the Government of Western Austalia'a Exploration Incentive Scheme (EIS). The drillhole is located on Sturt Meadows Station near the northern edge of the Bannockburn open pit, about 5.7 km north-northeast of Six Mile Well, 4.8 km east-northeast of Outcamp Well, and 4.8 km south-southeast of North Well mine.

Geological context

The unit sampled is a porphyritic microgranite near the western margin of the Mount Clifford greenstone belt in the Kalgoorlie Terrane (Williams and Oversby, 1996). The porphyritic microgranite sampled in drillcore BBDD0002 is about 3.2 m thick and has weakly sheared contacts with the metakomatiite host rock it intrudes (Urquhart et al., 2019). A metadacite collected about 25 km to the east yielded an igneous crystallization age of 2692 ± 4 Ma (GSWA 112159, Nelson, 1995).

Petrographic description

The sample is a crystal-rich porphyritic microgranite (Fig. 1) of rhyolitic or dacitic composition, consisting of about 60% groundmass, 30–35% plagioclase phenocrysts, 7% quartz phenocrysts, <1% sulfide minerals and trace titanite, apatite and zircon. The groundmass is a recrystallized mosaic of quartz and feldspar containing minor aligned laths of muscovite and chlorite up to 0.4 mm long that define a weak flow foliation, best developed adjacent to phenocrysts. Plagioclase is subhedral to euhedral, tabular, up to 5 mm long, has been albitized (compositions around An_6), and partially altered to muscovite and carbonate minerals. Quartz is anhedral, rounded, transparent, and up to 2 mm in size. Some quartz is strained, some has been completely recrystallized to a granoblastic mosaic of unstrained grains, and some are new grains. Although most plagioclase and quartz occurs as individual phenocrysts, some occur in glomeroporphyritic aggregates up to 5 mm in diameter. Sulfide minerals include irregular stringers of pyrrhotite up to 0.5 mm long and disseminated cubes of pyrite about 0.1 mm in size. Pale pink titanite is anhedral to subhedral and up to 0.3 mm in size. Bladed euhedral apatite laths are up to 0.15 mm long. The sample is not obviously deformed but has been recrystallized at low metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless and subhedral to euhedral. The crystals are up to 300 µm long, and elongate, with aspect ratios up to 9:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 24–25 June and 8–9 July 2021, using SHRIMP-B. Analyses 1.1 to 18.1 (spot numbers 1–20) were obtained during the first session, together with 14 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.83% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.30% (1σ). Analyses 19.1 to 34.1 (spot numbers 21–39) were obtained during the second session, together with eight analyses of the M257 standard, of which six analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.01% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.44% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Thirty-nine analyses were obtained from 34 zircons. Results are listed in Table 1, and shown in concordia diagrams (Figs 3, 4).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Twelve analyses indicate dates <2600 Ma and are >5% discordant. The dates obtained from these 12 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 27 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ and $^{238}U/^{206}Pb^*$ ratios.

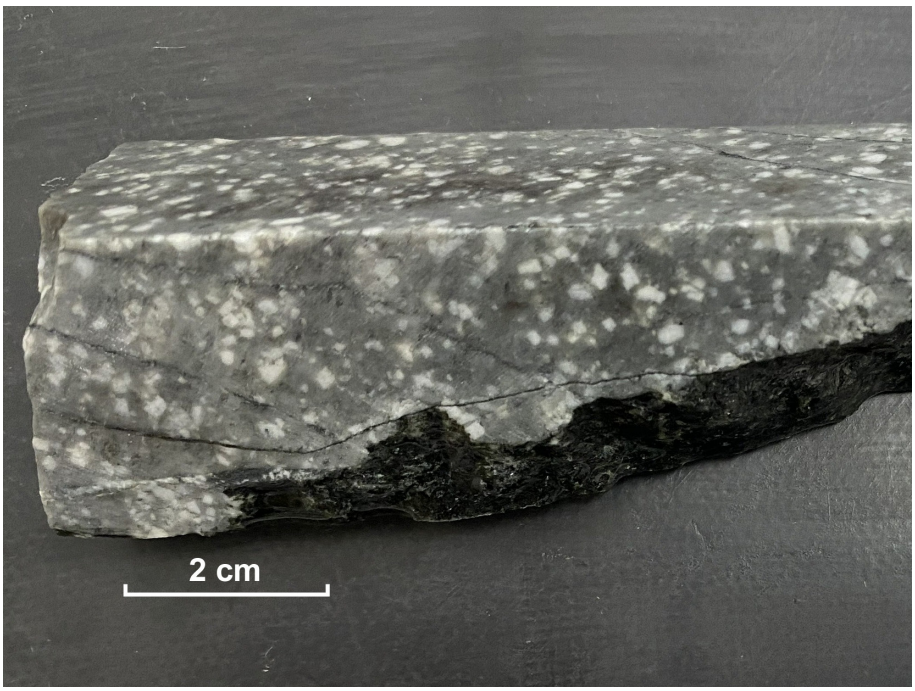

Figure 1. **Drillcore image (wet surface) of sample 249438: porphyritic microgranite, Bannockburn mine**

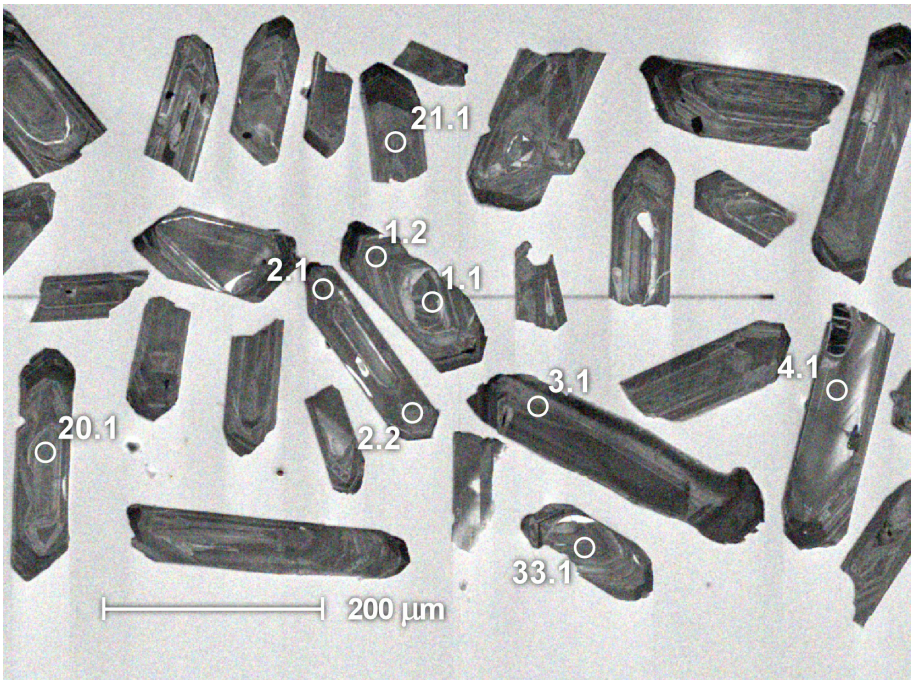

Figure 2. **Cathodoluminescence image of representative zircons from sample 249438: porphyritic microgranite, Bannockburn mine. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 249438: porphyritic microgranite, Bannockburn mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	35	30.1	521	157	0.30	0.003	2.194	0.026	0.17636	0.00047	2.194	0.026	0.17633	0.00047	2421	25	2619	4	7.5
I	21	19.1	436	127	0.29	0.008	2.099	0.025	0.17658	0.00053	2.099	0.025	0.17650	0.00053	2512	26	2620	5	4.1
I	22	20.1	312	59	0.19	0.075	2.113	0.026	0.17749	0.00062	2.115	0.026	0.17682	0.00064	2496	26	2623	6	4.8
I	14	12.1	128	71	0.56	0.274	2.154	0.027	0.18015	0.00106	2.160	0.027	0.17771	0.00122	2453	26	2632	11	6.8
I	6	5.1	228	11	0.05	0.142	2.087	0.024	0.17995	0.00084	2.090	0.024	0.17869	0.00091	2520	24	2641	8	4.6
I	32	17.2	171	114	0.67	0.071	2.079	0.027	0.17943	0.00079	2.081	0.027	0.17880	0.00082	2530	28	2642	8	4.2
I	31	27.1	227	65	0.29	0.015	2.044	0.026	0.17893	0.00073	2.044	0.026	0.17880	0.00073	2567	28	2642	7	2.8
I	11	9.1	88	28	0.32	0.067	1.965	0.028	0.17963	0.00124	1.967	0.028	0.17903	0.00129	2650	31	2644	12	-0.2
I	19	17.1	197	134	0.68	0.050	2.078	0.024	0.17964	0.00083	2.079	0.024	0.17920	0.00086	2532	25	2645	8	4.3
I	2	1.2	258	165	0.64	0.087	2.003	0.022	0.18039	0.00071	2.005	0.022	0.17961	0.00075	2608	24	2649	7	1.5
I	3	2.1	458	115	0.25	0.012	2.045	0.021	0.17980	0.00054	2.046	0.021	0.17969	0.00055	2566	22	2650	5	3.2
I	5	4.1	236	253	1.08	0.066	1.958	0.022	0.18032	0.00076	1.959	0.022	0.17973	0.00079	2658	25	2650	7	-0.3
I	9	7.1	259	50	0.20	0.096	2.004	0.022	0.18072	0.00076	2.006	0.022	0.17986	0.00080	2608	24	2652	7	1.7
I	26	22.1	433	109	0.25	0.021	2.025	0.024	0.18021	0.00050	2.025	0.024	0.18002	0.00051	2587	26	2653	5	2.5
I	34	29.1	325	83	0.26	0.034	2.077	0.025	0.18055	0.00060	2.078	0.026	0.18025	0.00061	2533	26	2655	6	4.6
I	15	13.1	212	50	0.23	-0.009	1.956	0.022	0.18026	0.00081	1.956	0.022	0.18034	0.00082	2662	25	2656	8	-0.2
I	17	15.1	273	60	0.22	0.028	2.006	0.022	0.18081	0.00071	2.007	0.022	0.18056	0.00072	2607	24	2658	7	1.9
I	23	2.2	441	104	0.24	0.004	2.013	0.024	0.18109	0.00050	2.013	0.024	0.18106	0.00051	2600	26	2663	5	2.4
I	13	11.1	290	58	0.20	0.034	1.980	0.022	0.18143	0.00069	1.980	0.022	0.18113	0.00071	2635	24	2663	6	1.1
I	16	14.1	341	90	0.26	0.017	1.987	0.021	0.18154	0.00063	1.988	0.021	0.18139	0.00064	2627	23	2666	6	1.4
I	39	34.1	302	111	0.37	0.000	2.004	0.025	0.18143	0.00062	2.004	0.025	0.18143	0.00062	2610	27	2666	6	2.1

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	ƒ204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
X	29	25.1	252	58	0.23	0.019	2.150	0.027	0.18152	0.00068	2.150	0.027	0.18134	0.00069	2462	26	2665	6	7.6
X	38	33.1	260	147	0.57	0.006	2.136	0.027	0.18218	0.00067	2.136	0.027	0.18212	0.00068	2475	26	2672	6	7.4
X	10	8.1	307	214	0.70	0.025	1.992	0.022	0.18429	0.00068	1.992	0.022	0.18406	0.00069	2622	24	2690	6	2.5
X	1	1.1	305	194	0.64	0.037	1.942	0.021	0.18476	0.00068	1.943	0.021	0.18443	0.00069	2677	24	2693	6	0.6
X	25	8.2	206	101	0.49	0.037	1.983	0.025	0.18497	0.00074	1.984	0.025	0.18463	0.00075	2631	28	2695	7	2.4
X	18	16.1	65	36	0.56	0.352	1.837	0.028	0.19650	0.00148	1.843	0.028	0.19336	0.00172	2794	35	2771	15	-0.8
D	12	10.1	1934	98	0.05	0.012	3.436	0.054	0.12110	0.00083	3.437	0.054	0.12099	0.00084	1646	23	1971	12	16.5
D	24	21.1	882	396	0.45	0.064	2.868	0.041	0.13552	0.00075	2.870	0.041	0.13495	0.00076	1927	24	2163	10	10.9
D	4	3.1	865	73	0.08	0.113	2.665	0.043	0.15134	0.00098	2.668	0.043	0.15033	0.00099	2052	28	2350	11	12.7
D	27	23.1	627	245	0.39	0.062	2.590	0.030	0.15098	0.00040	2.592	0.030	0.15043	0.00042	2104	22	2351	5	10.5
D	7	5.2	635	178	0.28	0.016	2.748	0.027	0.15286	0.00047	2.748	0.027	0.15271	0.00047	2001	17	2377	5	15.8
D	33	28.1	501	187	0.37	0.044	2.616	0.031	0.15471	0.00045	2.617	0.031	0.15432	0.00047	2086	22	2394	5	12.9
D	8	6.1	480	83	0.17	0.114	2.374	0.034	0.15615	0.00156	2.377	0.035	0.15514	0.00157	2264	28	2403	17	5.8
D	37	32.1	570	8	0.01	0.060	2.464	0.029	0.16181	0.00045	2.466	0.029	0.16127	0.00047	2195	23	2469	5	11.1
D	30	26.1	657	200	0.30	0.031	2.545	0.030	0.16214	0.00041	2.546	0.030	0.16186	0.00042	2136	22	2475	4	13.7
D	20	18.1	610	224	0.37	0.016	2.547	0.026	0.16269	0.00050	2.547	0.026	0.16254	0.00050	2135	19	2482	5	14.0
D	36	" 31.1"	621	196	0.32	0.018	2.570	0.030	0.16325	0.00043	2.570	0.030	0.16309	0.00044	2118	22	2488	4	14.9
D	28	" 24.1"	464	146	0.31	0.004	2.414	0.029	0.16441	0.00049	2.414	0.029	0.16437	0.00049	2234	23	2501	5	10.7

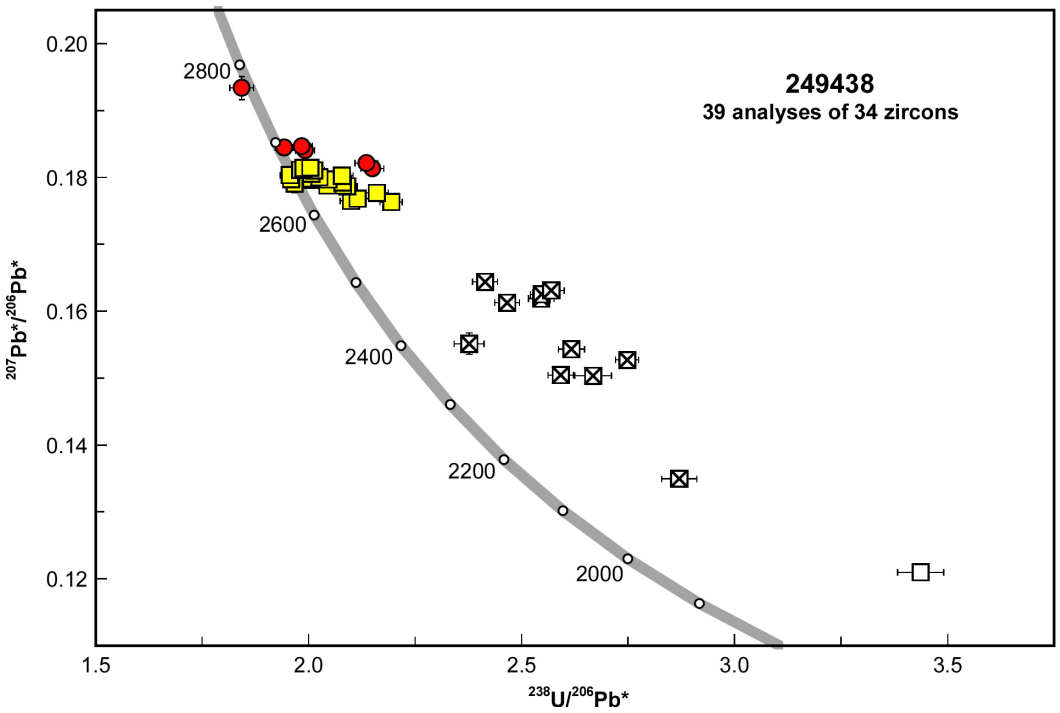

Figure 3. U–Pb analytical data for sample 249438: porphyritic microgranite, Bannockburn mine. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (inherited zircon cores); crossed squares indicate Group D (date <2600 Ma and discordance >5%)

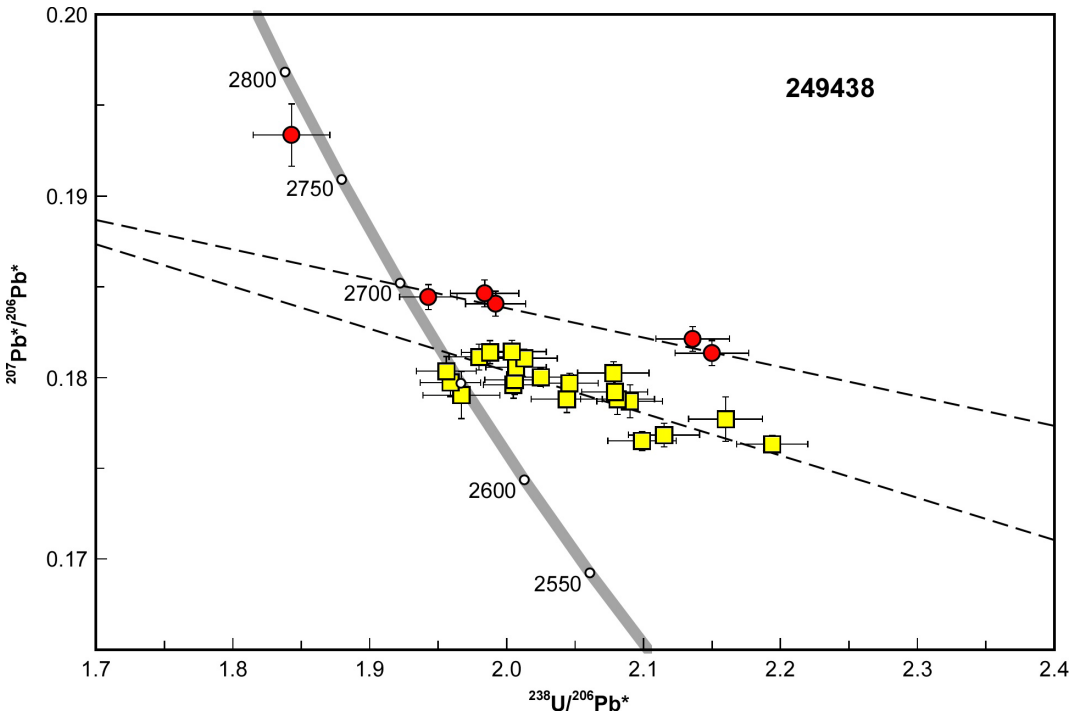

Figure 4. Expanded view of U–Pb analytical data for sample 249438: porphyritic microgranite, Bannockburn mine. Dashed lines indicate discordia regressions through data in Groups I and X; other symbols as in Figure 3

Group I comprises 21 analyses of 19 zircons (Table 1), which yield a discordia with intercepts at 2669 ± 8 and c. 900 Ma (MSWD = 1.2).

Group X comprises six analyses of five zircon cores (Table 1), which yield $^{207}Pb^{*}/^{206}Pb^{*}$ dates of 2771–2665 Ma. Five analyses of four zircon cores yield a discordia with intercepts at 2699 ± 12 and c. 600 Ma (MSWD = 0.26).

The date of 2669 ± 8 Ma for the 21 analyses in Group I is interpreted as the igneous crystallization age of the microgranite. The dates of 2771–2665 Ma for the six analyses in Group X are interpreted as the ages of inherited zircons. The date of 2699 ± 12 Ma for five analyses in Group X is interpreted as age of an inherited component.

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Lowrey, JR 2021, 249438: porphyritic microgranite, Bannockburn mine; Geochronology Record 1839: Geological Survey of Western Australia, 6p.

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